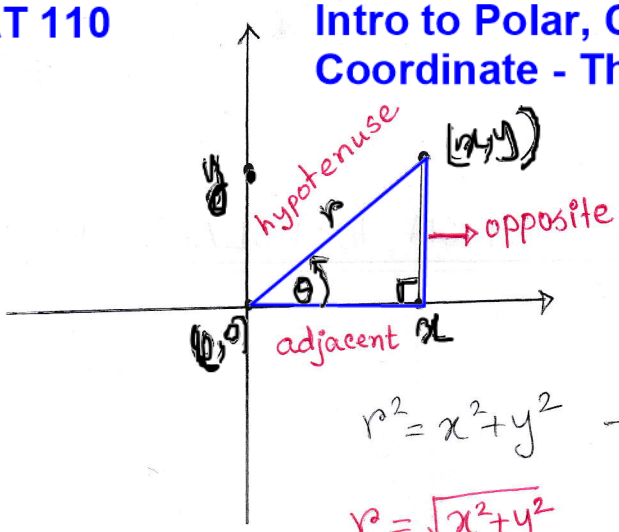
WEEK 10 **MAT 110**

Intro to Polar, Cylindrical & Spherical Coordinate - Theorem

PART A



$r^2 = x^2 + y^2 \longrightarrow$ by Pythagorean Thm

$r = \sqrt{x^2+y^2}$

$x = r\sin\theta \longrightarrow \sin\theta = \frac{y}{r} \longrightarrow \operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{r}{y}$

$y = r\cos\theta \longrightarrow \cos\theta = \frac{x}{r} \longrightarrow \sec\theta = \frac{1}{\cos\theta} = \frac{r}{x}$

$\tan\theta = \frac{y}{x} \longrightarrow \cot\theta = \frac{1}{\tan\theta} = \frac{x}{y}$

$x = r\cos\theta$ (i) , $y = r\sin\theta$ (ii)

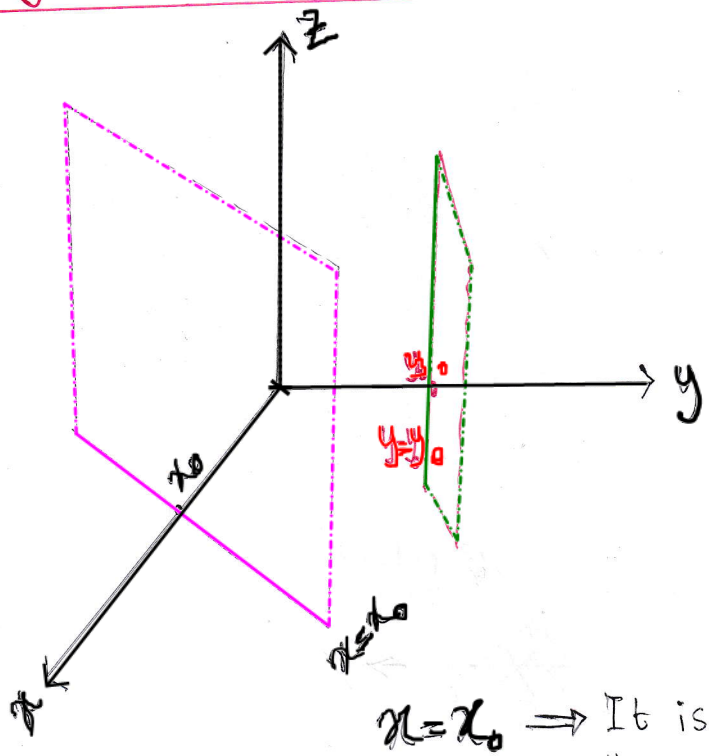
Sq eqn (i) $\rightarrow x^2 = r^2\cos^2\theta$

Sq eqn (ii) $\rightarrow y^2 = r^2\sin^2\theta$

Add $\rightarrow x^2+y^2 = r^2(\cos^2\theta+\sin^2\theta)$

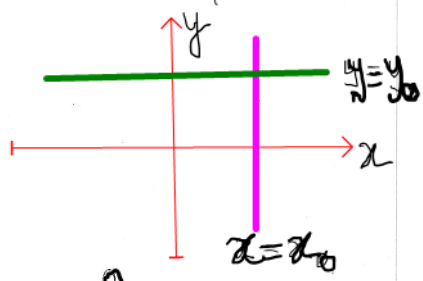
$r^2 = x^2+y^2$

Cylindrical Coordinates



3-D Space

$\underbrace{P(r, \theta, z)}_{\text{Coordinate point}}$



$x = x_0 \Rightarrow$ It is a plane through x-axis $\Big\}$ 3-D

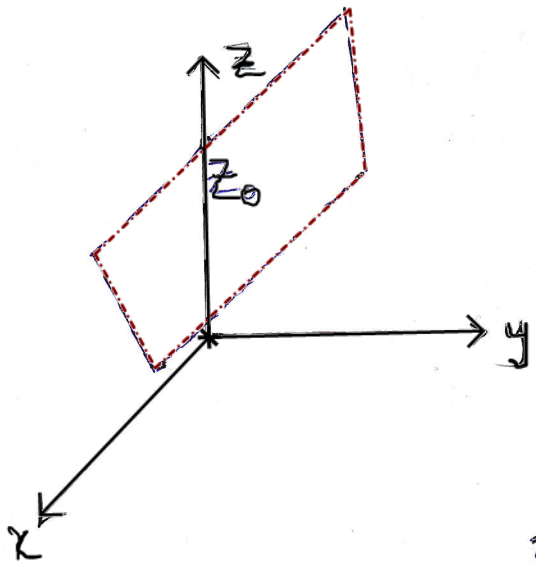
$x = x_0 \Rightarrow$ a $\underbrace{\text{straight}}_{\text{(vertical)}}$ line $\Big\}$ 2-D

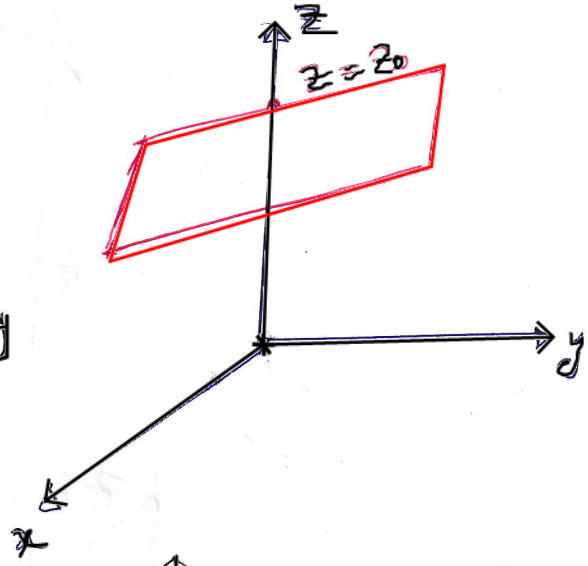
$x = x_0 \Rightarrow$ This plane can be extended infinitely in the direction of y-axis & in the direction of z-axis.

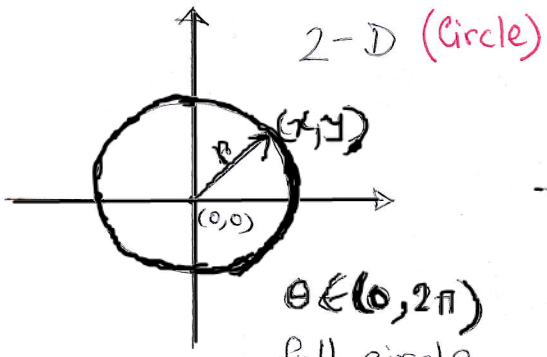
Similarly $y = y_0 \Rightarrow$ A plane through y-axis $\rightarrow$ 3-D
$\Rightarrow$ A horizonal line $\rightarrow$ 2-D

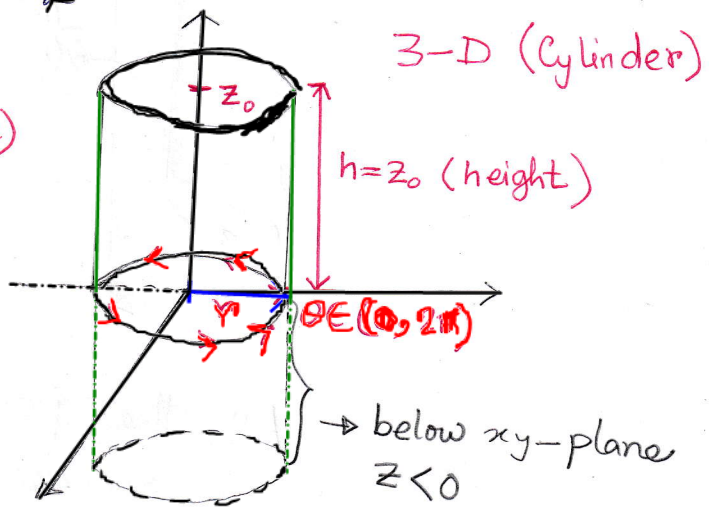
$y = y_0 \Rightarrow$ This plane can be extended infinitely in the direction of x-axis & in the direction of z-axis.

$z = z_0 \Rightarrow$ A plane parallel to the horizon or parallel to xy-plane.

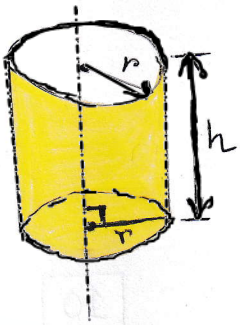


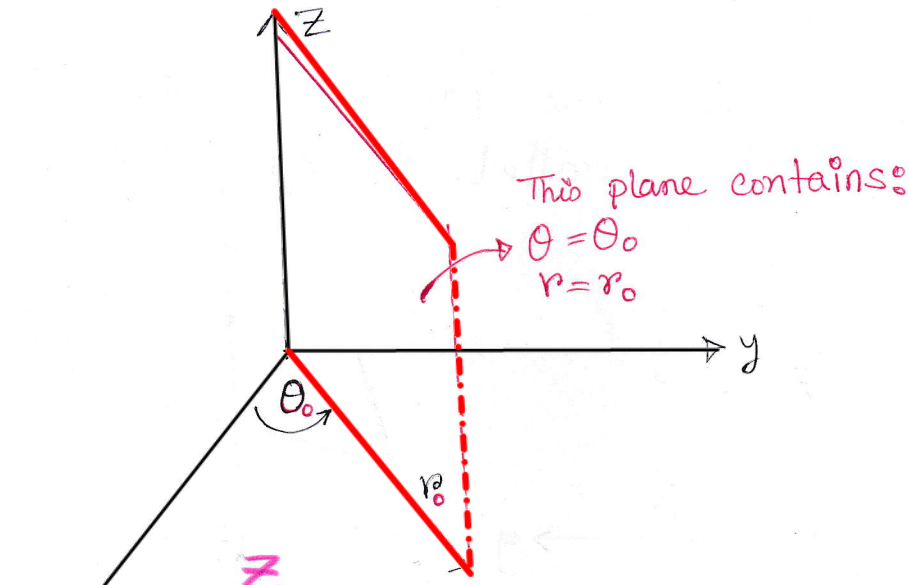


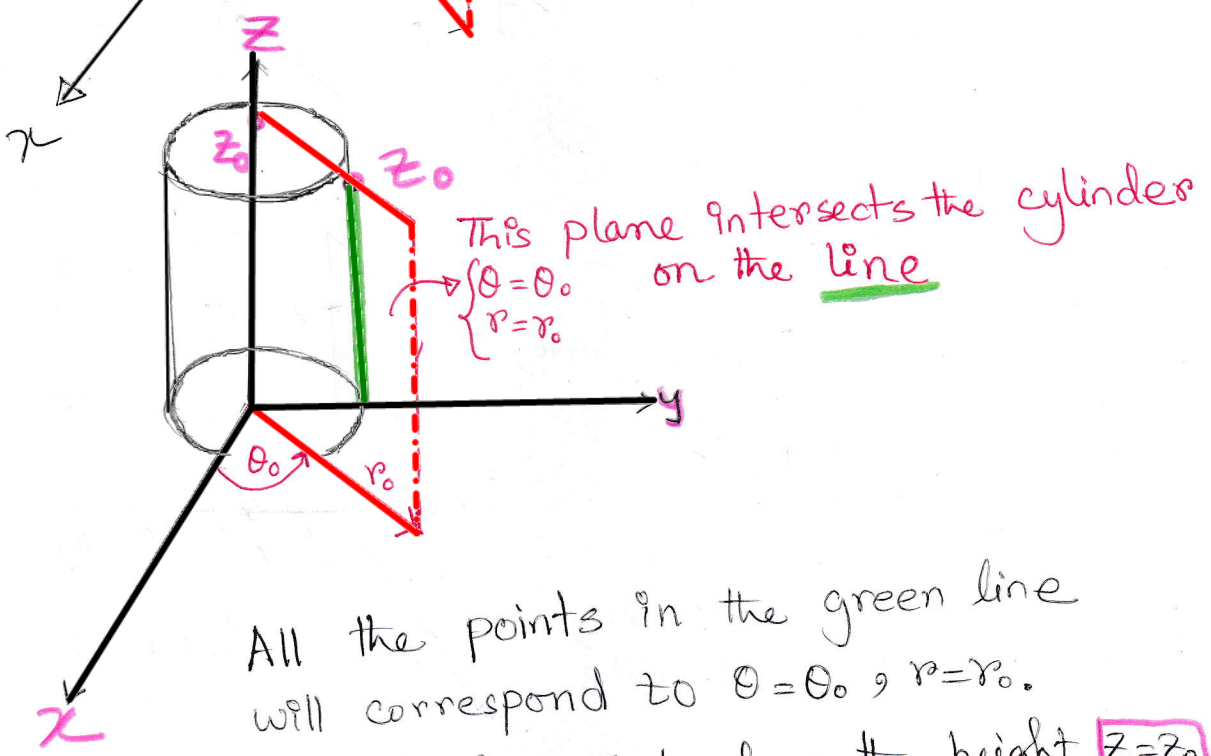




A right circular cylinder is a cylinder that has a closed circular surface having two parallel bases on both ends and whose elements are perpendicular $\perp$ to its base. It is also called a right cylinder.



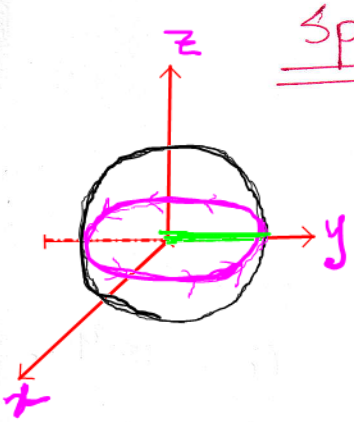


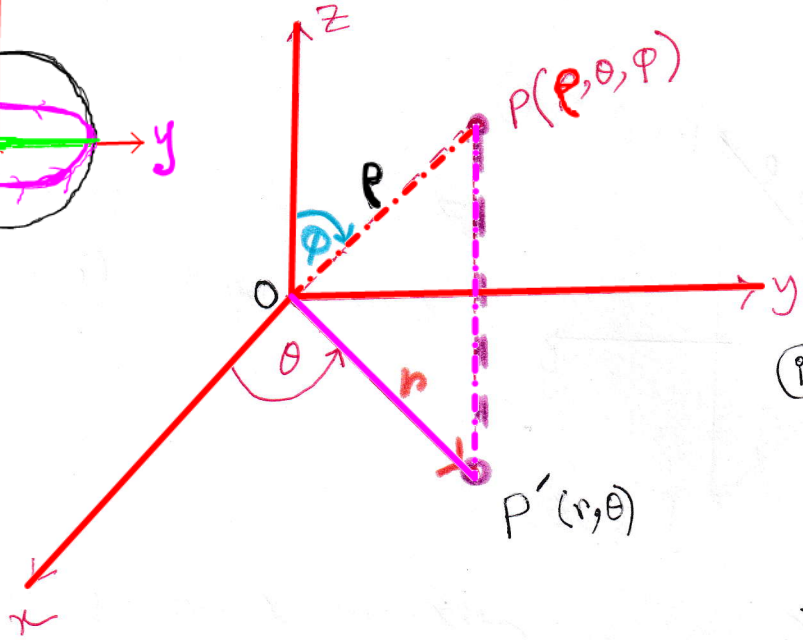


All the points in the green line will correspond to $\theta = \theta_0$, $r = r_0$.
Similarly if we introduce the height $\boxed{z = z_0}$, it will correspond to the green line as well.

Rectangular to Cylindrical	Cylindrical to Rectangular
$x = r\cos\theta$	$r = \sqrt{x^2 + y^2}$
$y = r\sin\theta$	$\tan\theta = \frac{y}{x}$
$z = z$	$z = z$

Spherical Coordinate





Eqn of sphere, center (0,0,0)
$[x^2+y^2+z^2=\rho^2]$

(i) $\rho \to$ rho
$\rho = |\overrightarrow{OP}| \to$ radius of sphere

$\rho \geq 0$

(ii) $\theta \to$ angle from x axis to the projection of the point P on xy-plane

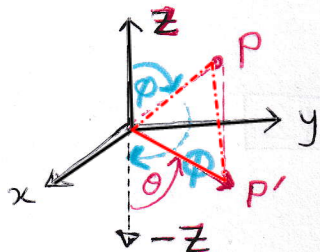
$\to$ Vertical projection of P on xy-plane

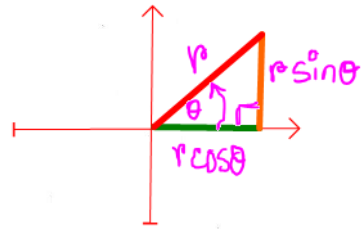
(iii) $\to$ Let's call the projecting point on xy-plane P' from the point P.

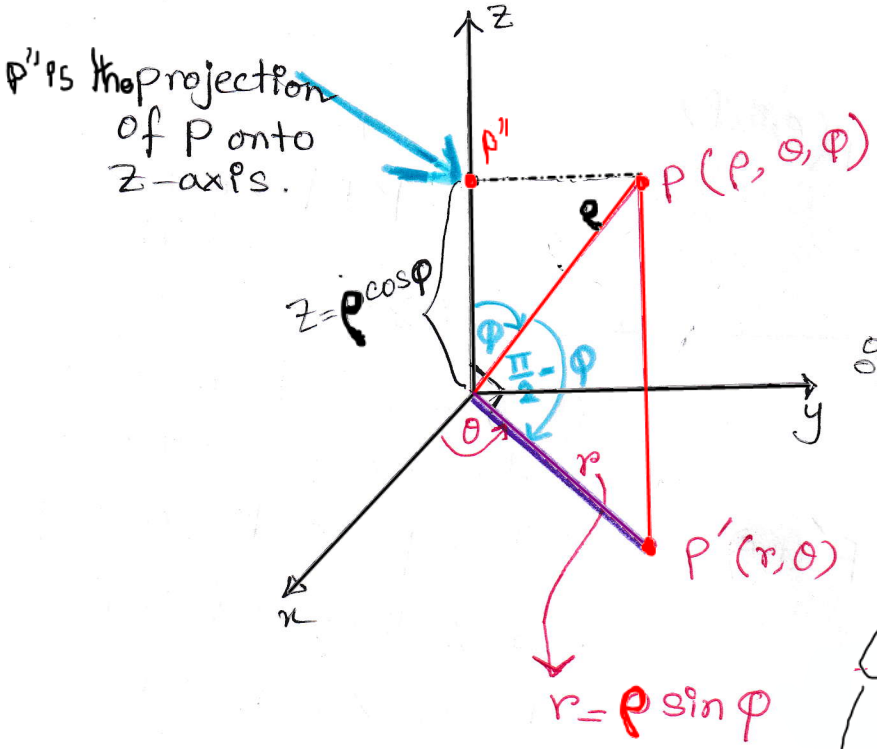
$P' \to (r,\theta)$ $\quad r = d\{(0,0), P'\}$

$\theta \in (0, 2\pi)$

(iv) $\varphi \to$ angle between OP to z-axis.
It starts to rotate from z-axis towards xy plane. $\therefore \varphi \in (0, \pi/2)$
Then it rotates from xy-plane towards the -ve side of z-axis. $\therefore \varphi \in (\pi/2, \pi)$
Hence the complete turn of φ is $(0, \pi)$.







$z \perp$ on xy-plane

$\therefore$ z axis to the line is $90°$.

We know

$x = r\cos\theta = \rho \sin\varphi \cos\theta$

$y = r\sin\theta = \rho \sin\varphi \sin\theta$

since $r = \rho\sin\varphi$

$z = \rho\cos\varphi$

Rectangular to Spherical:

$x = \rho \sin\varphi \cos\theta$

$y = \rho \sin\varphi \sin\theta$

$z = \rho \cos\varphi$

Spherical to Rectangular

$\rho^2 = r^2 + z^2$

$\rho^2 = x^2 + y^2 + z^2$

$\tan\theta = \dfrac{y}{x}$

$\cos\varphi = \dfrac{|z|}{\rho} = \dfrac{|z|}{\sqrt{x^2+y^2+z^2}}$

consider z as a +ve value hence take $|z|$ to evaluate φ as $\varphi \in (0, \pi)$